Zhengxiong Li

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Education

University of Wisconsin-Madison Madison, USA
Ph.D., Department of Electrical and Computer Engineering *09/2024 – 05/2029 (Expected)*

Wuhan University Wuhan, China
B.S., Microelectronics Science and Engineering, Hongyi Honor College *09/2020 – 06/2024*

Publications

- **Li, Z.**, Huang, T., Ogras, U. (2026). *CREDIT: Cost-guided Reduction-reuse with Efficient DSMEM Inter-CTA Tiling*. IEEE High Performance Extreme Computing Conference (HPEC), Virtual.
- **Li, Z.**, Huang, T., Ogras, U. (2026). *SET: Stream-Event-Triggered Scheduling for Efficient CUDA Graph Pipelines*. International European Conference on Parallel and Distributed Processing (Euro-Par), Pisa, Italy. [Preprint]
- Jie, T., **Li, Z.**, Ogras, U., Huang, T. (2025). *A Scalable Code Generation Flow for Heterogeneous Parallel RTL Simulation using MLIR*. IEEE High Performance Extreme Computing Conference (HPEC), Virtual. [DOI]
- Li, X.†, **Li, Z.**†, et al. (2024). *MSGM: An Advanced Deep Multi-Size Guiding Matching Network for Whole Slide Histopathology Images Addressing Staining Variation and Low Visibility Challenges*. IEEE Journal of Biomedical and Health Informatics (JBHI). [DOI] †Equal contribution.
- Li, X.†, Hu, T.†, **Li, Z.**†, et al. (2024). *MRRM: Advanced Biomarker Alignment in Multi-Staining Pathology Images via Multi-Scale Ring Rotation-Invariant Matching*. IEEE Journal of Biomedical and Health Informatics (JBHI). [DOI] †Equal contribution.
- Hu, M., Jiang, X., **Li, Z.**. (2023). *A Resource-efficient FIR Filter Design Based on an RAG Improved Algorithm*. International Conference on Circuits and Systems (ICCS). [DOI]

Honors & Awards

63rd Design Automation Conference (DAC) Young Fellow 07/2026
UW-Madison David and Sarah Epstein Fellowship 06/2026
62nd Design Automation Conference (DAC) Young Fellow 07/2025

Research Experience

CREDIT: Cost-guided Reduction-reuse with Efficient DSMEM Inter-CTA Tiling Madison, USA
Research Assistant; Supervisor: Prof. Tsung-Wei Huang and Prof. Umit Ogras *01/2026 – 06/2026*

- Quantitatively profiled the performance of distributed shared memory (DSMEM) in terms of read/write latency, throughput, and synchronization overhead.
- Developed a profile-guided cost model that compares the time we can save against extra overheads introduced by DSMEM, including cluster synchronization, DSMEM communication, and occupancy costs, to predict profitability.
- Evaluated diverse workloads on NVIDIA RTX 5090 and H100 against `torch.compile`, Triton, and optimized no-DSMEM baselines, and validated the off-chip traffic reduction with Nsight Compute.

SET: Stream-Event-Triggered Scheduling for Efficient CUDA Graph Pipelines Madison, USA
Research Assistant; Supervisor: Prof. Tsung-Wei Huang and Prof. Umit Ogras *06/2025 – 11/2025*

- Analytically decomposed scheduling overheads in CUDA graph pipelines into intra-batch and inter-batch components and explained why kernel gaps persist despite graph instantiation and multi-stream execution.
- Achieved adaptive task assignment with O(1) synchronization overhead while preserving per-job ordering and memory safety by using bounded in-flight CUDA graph executables and callback-driven coordination to minimize inter-kernel gaps.
- Evaluated on six representative compute- and memory-bound workloads on NVIDIA RTX 3090 and RTX 5090, and demonstrated 1.15–1.44$\times$ speedup and 18–54% lower scheduling overheads than state-of-the-art host-side scheduling baselines.

PROWESS: Processor Reconfiguration for Wideband Spectrum Sensing Madison, USA
Research Assistant; Supervisor: Prof. Umit Ogras *09/2024 – 03/2025*

- Implementing a system-level domain-specific SoC simulation framework (DS3) on Zynq platform. Porting the core on-chip scheduler module from Python to C to meet resource restrictions on embedded systems. Finding the optimal resource configuration that can guide the hardware team's design.

- Studied the communication overheads between on-chip scheduler (PS side) and program launcher (PL side) through AXI protocols. Found the optimal job distribution for task scheduling to achieve best efficiency and performance.
- Optimized the DS3 simulator to enable massive parallelization, including decoupling improper program design and refactoring stale data structures. Achieved up to 5x speedup.
- Performed design space exploration based on DS3 simulator and analyzed the generated data. Studied the impact on throughput caused by different factors, including processing elements (PEs) mesh architecture, iMEM size, communication overheads, etc. Identified the bottleneck to achieving higher throughput gains.

Development of an FPGA-Based Automatic Calibration System for RFIC Wuhan, China
Undergraduate Researcher; Supervisor: Prof. Xianyang Jiang *05/2022 – 04/2023*

- Designed a pair of SPI protocol communication interfaces to enable data transmission among the PC, FPGA board, and target chip to be calibrated.
- Employed the DDS method to develop a system generating periodically vibrating waveforms with high stability.
- Developed a series of FIR filters and verified their efficacy using Matlab while integrating various decimation modules to filter the digital data acquired from ADC for precise signal processing.
- Conducted simulation and debugging procedures for FPGA and PCB boards, leading to their successful deployment.

FPGA/GPU-Based Numerical Solvers for Sustainable Energy System Simulation Edmonton, Canada
Research Intern; Mitacs Internship; Supervisor: Prof. Venkata Dinavahi *06/2023 – 09/2023*

- Developed a high-performance solution for real-time simulation of electrical systems, ensuring the hardware's ability to predict the next 1us' situation within a strict time frame, such as 100ns or shorter.
- Trained GRU and LSTM models for Synchronous Machine and Induction Machine modeling.
- Implemented the pre-trained model on Xilinx Data Center Acceleration Card (U250), aiming to achieve real-time or faster than real-time simulation for the entire electrical system.

Coursework Projects

GPU Benchmarks Porting and Simulation for AMD GPUs on gem5 Madison, USA
Project Leader *09/2024 – 12/2024*

- Ported the Polybench and Lonestar benchmark suites from CUDA to AMD HIP, manually resolving inline assembly and library mismatches to enable cross-platform compatibility.
- Simulated AMD Vega 20 microarchitecture using the gem5 simulator to analyze performance characteristics, validating simulation accuracy against real-world Radeon VII hardware execution.
- Investigated the impact of dynamic vs. static register allocation schemes on irregular graph algorithms (BFS, SSSP), identifying resource contention bottlenecks and potential logic errors within the simulator's scheduling model.

FPGA-Based Ultra-High-Speed Panoramic Camera Image Stitching Project Wuhan, China
Core Member *04/2023 – 06/2023*

- Improved the performance of existing software algorithm and transformed it into hardware description language.
- Implemented the improved algorithm on an Xilinx FPGA development board to achieve high resolution and high frame rate for panoramic camera image stitching.
- Collaborated with a diverse team, including fellow students from the laboratory, engineers from the company, and students from Huazhong University of Science and Technology.

Academic Services

Artifact Evaluation Committee, CASES 2026
Teaching Assistant, Digital Logic Circuit Lab 2023

Skills & Tools

Programming: CUDA, C & C++, Python, LLVM, MLIR, Verilog, Matlab
Laboratory Techniques: Nsight Systems Profiler, GDB, Vitis, Vivado, Quartus, Modelsim, FPGA hardware debug, FPGA software development
Languages: Mandarin (Native), English (Fluent)